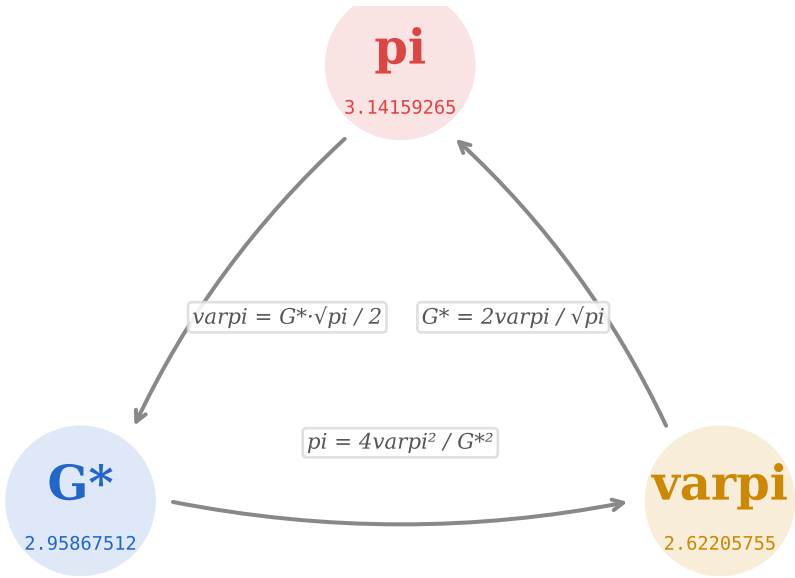
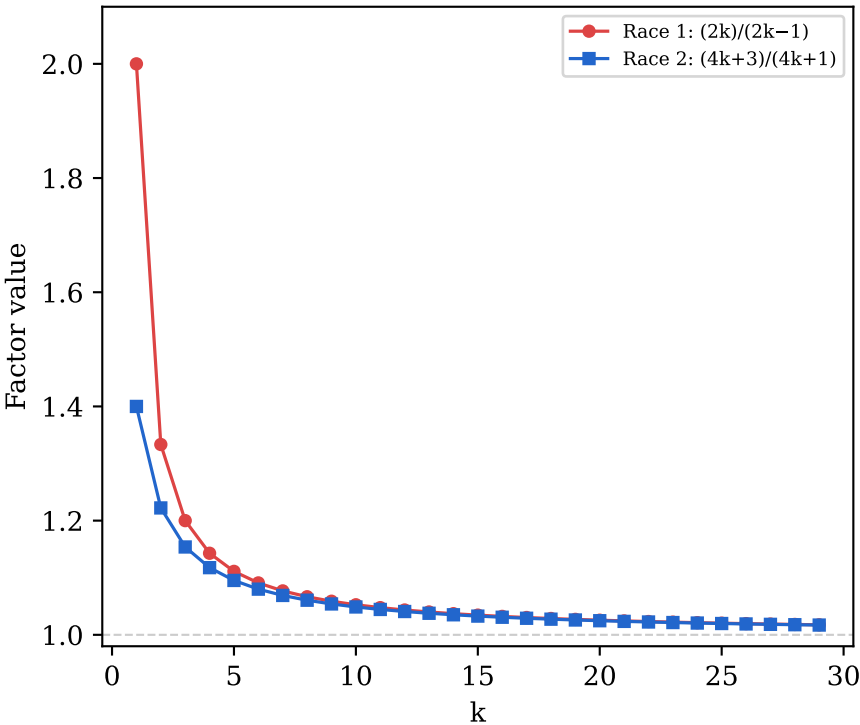


# The Triad: pi, G\*, and varpi — Three Constants, One Identity

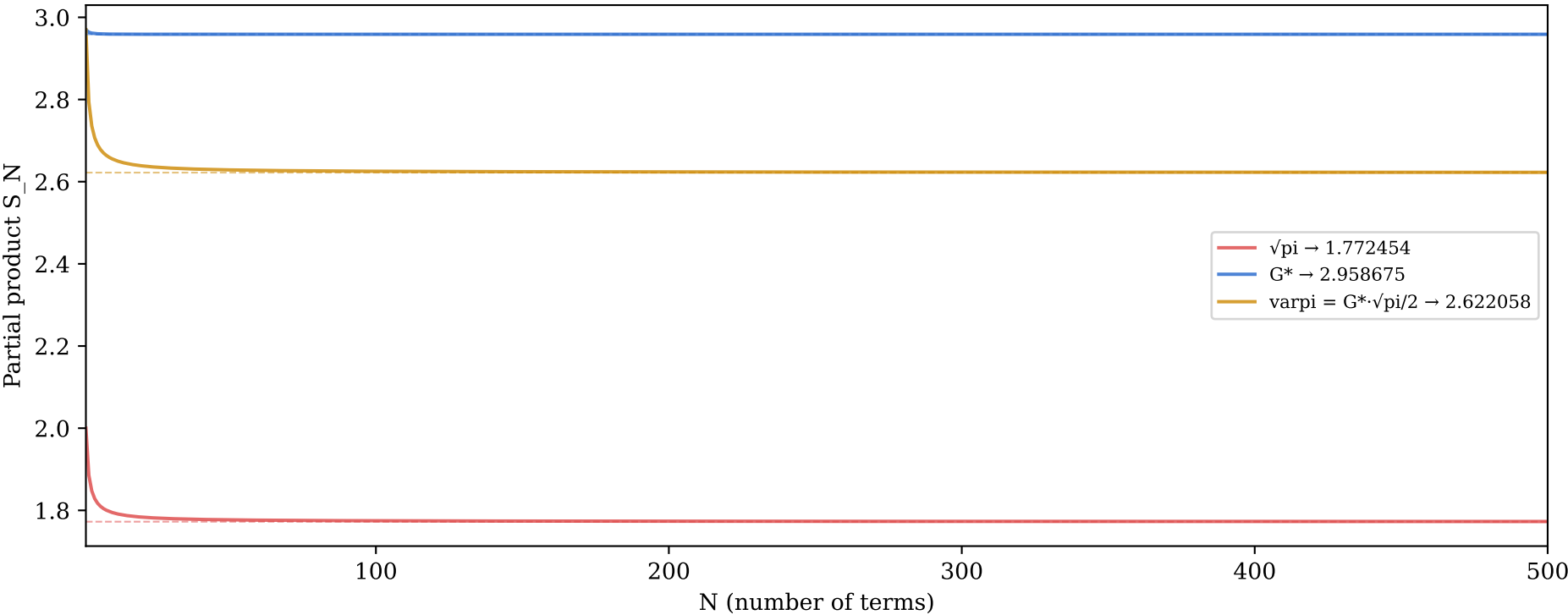
The Triad Identity



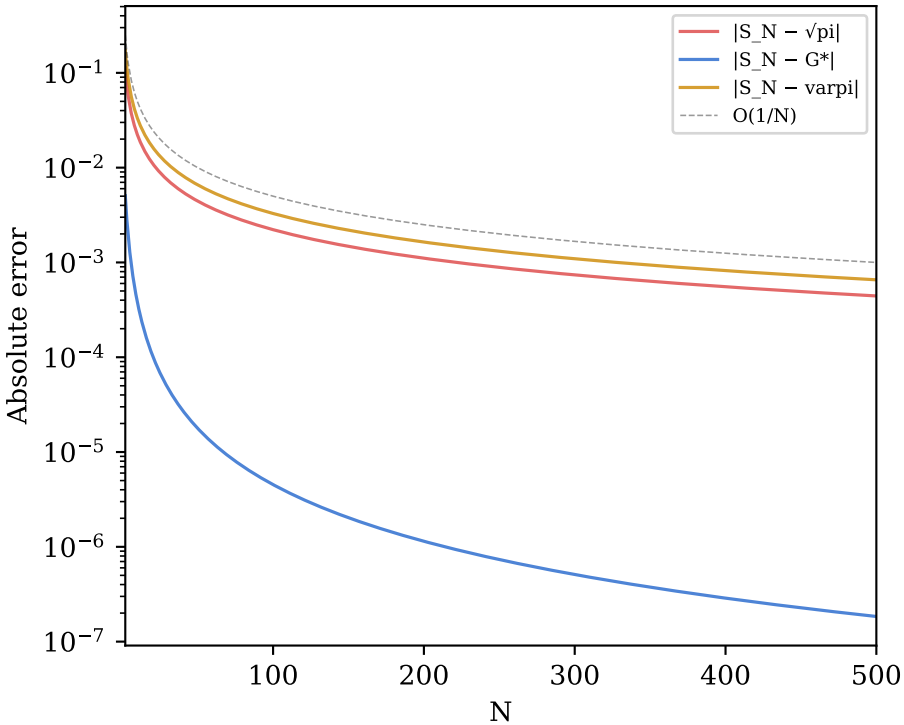
Individual Wallis Factors



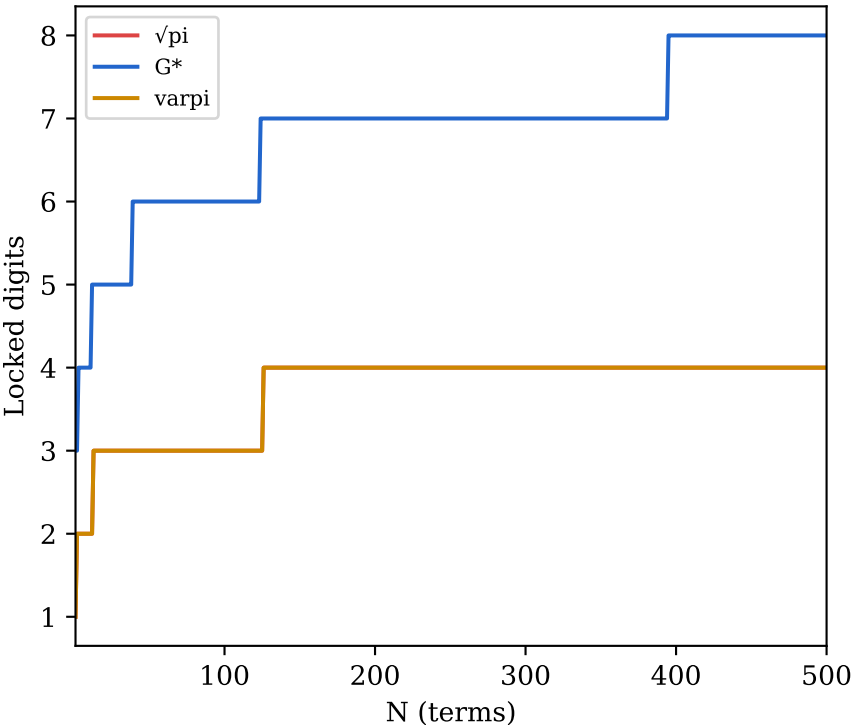
Convergence of the Two Races + Composite varpi



Error Decay (log scale)



Correct Digits vs. Terms



Three Merge Directions

**pi + G\* → varpi**

$$\text{varpi} = G^* \cdot \sqrt{\pi} / 2 = 2.6220575543$$

**varpi + G\* → pi**

$$\pi = 4 \text{varpi}^2 / G^{*2} = 3.1415926536$$

**pi + varpi → G\***

$$G^* = 2 \text{varpi} / \sqrt{\pi} = 2.9586751192$$

The Race Structure

**Race 1 (mod 2)**

$$\prod (2k)/(2k-1) \rightarrow \sqrt{\pi}$$

**Race 2 (mod 4)**

$$\prod (4k+3)/(4k+1) \rightarrow G^*$$

**Both races / 2**

$$\text{Race 1} \times \text{Race 2} / 2 \rightarrow \text{varpi}$$

**Vocabulary**

$$\text{evens vs odds} = \sqrt{\pi}$$

**Grammar**

$$\text{inert vs split} = G^*$$

**Sentence**

$$\text{vocabulary} \times \text{grammar} = \text{varpi}$$